

Program : Diploma in Instrumentation Engineering	
Course Code : 5083C	Course Title: Robotics and Control
Semester : 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To understand concepts, parts of robots and types of robots.
- To familiarize with the various drive systems for robot, sensors and their applications in robotic automation.
- To discuss about the various applications of robots, justification and implementation of robot.
- To conceptualize automation and understand applications of robots in various industries.

Course Prerequisites:

Topic	Course code	Course name	Semester
Electrical drives		Fundamentals of Electrical & Electronics	2
Sensors		Sensors and Transducers	3
Sensors		Process Variable Measurements	4

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the robot anatomy, classification, characteristics of robot.	15	Understanding

CO2	Explain the various robotic actuators on hydraulic, pneumatic and electrical drives.	17	Applying
CO3	Explain about various types of sensors and robot vision system.	16	Understanding
CO4	Explain the various applications of robots.	10	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	3						
CO3	3						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the robot anatomy, classification, characteristics of robot, advantages and disadvantages.		
M1.01	Explain the fundamentals of Robotics & laws of robotics.	2	Understanding
M1.02	Describe the Robot anatomy	5	Understanding
M1.03	Explain the classification of robots	6	Understanding
M1.04	List the advantages and disadvantages of robots	2	Understanding
Contents: Fundamentals of Robotics: Introduction; Degree of freedom, Asimov's laws of robotics, Definition; Robot anatomy (parts) and its working; Robot Components: Manipulator, End effectors; Construction of links, Types of joints; Classification of robots; Cartesian, Cylindrical, Spherical, Scara, Vertical articulated; Robot work Volumes, Comparison; Advantages and disadvantages of robots.			

CO2	Explain the various robotic actuators on hydraulic, pneumatic and electrical drives.		
M2.01	Explain different Robotic Drive Systems	5	Understanding
M2.02	Explain different actuation methods	4	Understanding
M2.03	Conversion between linear and rotary motion	4	Applying
M2.04	Explain the feedback devices	4	Understanding
	Series Test - I	1	
Contents: Robotic Drive System: Actuators; Hydraulic, Pneumatic and Electrical drives; Linear actuator; Rotary drives; AC servo motor; DC servo motors and Stepper motors; Conversion between linear and rotary motion; Feedback devices; Potentiometers; Optical encoders; DC tachometers;			
CO3	Explain about various types of sensors and concepts on robot vision system.		
M3.01	Explain the requirements of sensors	4	Understanding
M3.02	Explain the application of different sensors	4	Understanding
M3.03	Explain the robot's machine vision system	5	Understanding
M3.04	Applications of robot vision systems.	3	Understanding
Contents: Sensors: Requirements of a sensor; Principles and Applications of the following types of sensors: Position sensors (Encoders, Resolvers, Piezo Electric); Range sensors; Proximity sensing; Force and torque sensing, Tactile sensors. Introduction to Machine Vision: Robot vision system (scanning and digitizing image data); Cameras (Acquisition of images); Vidicon camera (Working principle & construction); Applications of Robot vision system: Inspection, Identification, Navigation & serving.			
CO4	Explain the various applications of robots.		
M4.01	Describe the elements of automated system	4	Understanding
M4.02	Explain the advanced automation functions	3	Understanding
M4.03	Explain the applications of robotics in various industries.	3	Understanding
	Series Test - II	1	

Contents:

Automation: Basic elements of automated system, advanced automation functions, levels of automation. Industrial Applications: Application of robots in machining; welding; assembly, material handling, biomedical applications.

Text / Reference:

T/R	Book Title/Author
R1	Mikell P. Weiss G.M., Nagel R.N., Odraj N.G., Industrial Robotics, McGraw Hill Singapore, 1996
R2	Ghosh, Control in Robotics and Automation: Sensor Based Integration, Allied Publishers, Chennai, 1998.
R3	Saeed B. Niku, - Introduction to Robotics: Analysis, Systems, Applications – Pearson Education Inc. New Delhi 2006.
R4	M.P. Groover, - Industrial Robotics: Technology, Programming and Applications –Tata McGraw Hill Co, 2001.
R5	Fu.K.S. Gonzalz.R.C and Lee C.S.G - Robotics Control, Sensing, Vision and Intelligence –, McGraw Hill Book Co, 1987.
R6	Yoram Koren - Robotics for Engineers –McGraw Hill Book Co, 1992.
R7	Ganesh S. Hedge, - A Text book on Industrial Robotics –Laxmi Publications Pvt. Ltd., New Delhi, 2008.
R8	S.R. Deb & Sankha Deb - Robotics Technology and Flexible Automation – Tata McGraw-Hill, 2010.
R9	Elements of Robotics Process Automation, Mukherjee, Khanna Publishing House, Delhi, 2018
R10	Issac Asimov I Robot, Ballantine Books, New York, 1986.
R11	Thomas R. Kurfess Ph.D., P.E. Robotics And Automation Handbook CRC press

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/courses/112/101/112101098/
2	http://www.nptelvideos.in/2012/12/robotics.html
3	https://www.guru99.com/robotic-process-automation-tutorial.html